

Zixuan Yuan

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FinTech Thrust & AI Thrust
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<https://hkust-finai-lab.github.io>

RESEARCH INTERESTS

Financial Text Analytics, LLM Reasoning, Agentic Orchestration, Cost-effective Vertical RAG, and Model Interpretability

EDUCATION

Ph.D., Information Technology Sep. 2018 - Jun. 2023
Rutgers Business School, Rutgers University
Advisor: Professor Hui Xiong

M.S., Information Technology Sep. 2015 - Jan. 2017
Rutgers Business School, Rutgers University

B.S., Economics Sep. 2011 - Jun. 2015
School of Economics & Management, Tongji University

SELECTED PUBLICATIONS¹

(* Equal Contribution, † Corresponding Author)

Referred Journal Publications

1. **Zixuan Yuan**, Junming Liu, Haoyi Zhou, Denghui Zhang, Hao Liu, Nengjun Zhu, and Hui Xiong†. LEVER: Online Adaptive Sequence Learning Framework for High-Frequency Trading. *IEEE Transactions on Knowledge and Data Engineering (TKDE, CCF-A, SEM-A)*, 2023.
2. Denghui Zhang, Yanchi Liu, **Zixuan Yuan**, Yanjie Fu, Haifeng Chen, and Hui Xiong†. Multi-Faceted Knowledge-Driven Pre-Training for Product Representation Learning. *IEEE Transactions on Knowledge and Data Engineering (TKDE, CCF-A, SEM-A)*, 2022.
3. Zekai Chen, Dingshuo Chen, Xiao Zhang, **Zixuan Yuan**, and Xiuzhen Cheng. Learning Graph Structures with Transformer for Multivariate Time Series Anomaly Detection in IOT. *IEEE Internet of Things Journal (IOTJ, SCI, IF=9.936)*, 2021.

Peer-Reviewed Conference Papers

1. Jian Chen*, Siyuan Li*, Chucheng Wan, **Zixuan Yuan**†. From Statute to Control Flow: Span-Grounded Deontic Trees for Defeasible Scope Parsing. The 32st ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD 2026, CCF-A**).
2. Junren Xiao, Ruiyao Miao, **Zixuan Yuan**†, Hui Xiong†. Beyond Black Boxes: An Energy-Based Unified Framework for Interpretable Stock Selection. The 32st ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD 2026, CCF-A**).
3. Meishu PENG*, Ziyue ZHANG*, Yi Zhang, Pengyang Wang, **Zixuan Yuan**†, Denghui Zhang†. Accelerating LLM Fine-Tuning via Embedding Knowledge Transfer. Findings of the Association for Computational Linguistics (**ACL findings 2026, CCF-A**).
4. Siyuan Li, Jian Chen, Rui Yao, Xuming Hu, Peilin Zhou, Weihua Qiu, Simin Zhang, Chucheng Dong, Zhiyao Li, Qipeng Xie, **Zixuan Yuan**†. Compliance-to-Code: Enhancing Financial

¹A more up-to-date list of publications is available at my homepage <https://hkust-finai-lab.github.io/>

Compliance Checking via Code Generation. The 31st ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD 2025, CCF-A**).

5. Jian Chen, Zhenyan Chen, Xuming Hu, Peilin Zhou, Yining Hua, Han Fang, Cissy Hing Yee Choy, Xinmei Ke, Jingfeng Luo, **Zixuan Yuan**[†]. DeKeyNLU: Enhancing Natural Language to SQL Generation through Task Decomposition and Keyword Extraction. Conference on Empirical Methods in Natural Language Processing (**EMNLP 2025, CCF-B**).
6. Jian Chen, Peilin Zhou, Yining Hua, Dading Chong, Meng Cao, Yaowei Li, Wei Chen, Bing Zhu, Junwei Liang[†], **Zixuan Yuan**[†]. ClimateIQA: A New Dataset and Benchmark to Advance Vision-Language Models in Meteorology Anomalies Analysis. The 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2025, CCF-A**).
7. Kaichen Zhang, **Zixuan Yuan**, Hui Xiong[†]. The Impact of Generative Artificial Intelligence on Market Equilibrium: Evidence from a Natural Experiment. The 20th Conference on Web and Internet Economics (**WINE 2024, CCF-A**).
8. Wenyuan Jiang*, Wenwei Wu*, Le Zhang*, **Zixuan Yuan**[†], Jian Xiang, Jingbo Zhou, Hui Xiong[†]. Killing Two Birds with One Stone: Cross-modal Reinforced Prompting for Graph and Language Tasks. The 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2024, CCF-A**).
9. Kaichen Zhang, **Zixuan Yuan**, Hui Xiong[†]. Optimized Cost Per Click in Online Advertising: A Theoretical Analysis. The 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD 2024, CCF-A**).
10. Nengjun Zhu, Jieyun Huang, Jian Cao, Liang Hu, **Zixuan Yuan**, Huanjing Gao. R2V-MIF: Rule-to-Vector Contrastive Learning and Multi-channel Information Fusion for Therapy Recommendation. The 33rd International Joint Conference on Artificial Intelligence (**IJCAI 2024, CCF-A**).
11. **Zixuan Yuan**, Hao Liu, Haoyi Zhou, Denghui Zhang, Xiao Zhang, Hao Wang, Hui Xiong[†]. Self-Paced Unified Representation Learning for Hierarchical Multi-Label Classification. In Proceedings of the 38th AAAI Conference on Artificial Intelligence (**AAAI 2024, CCF-A**).
12. **Zixuan Yuan**, Yada Zhu, Wei Zhang, Hui Xiong[†]. Earnings Call Analysis Using a Sparse Attention Based Encoder and Multi-Source Counterfactual Augmentation. The 4th ACM International Conference on AI in Finance (**ICAIF 2023**).
13. Denghui Zhang*, **Zixuan Yuan***, Hao Liu, Xiaodong Lin, and Hui Xiong[†]. Learning to Walk with Dual Agents for Knowledge Graph Reasoning. In Proceedings of the 36th AAAI Conference on Artificial Intelligence (**AAAI 2022, CCF-A**).
14. **Zixuan Yuan**, Hao Liu, Junming Liu, Yanchi Liu, Yang Yang, Renjun Hu, and Hui Xiong[†]. Incremental Spatio-Temporal Graph Learning for Online Query-POI Matching. In Proceedings of the 28th World Wide Web Conference (**WWW 2021, CCF-A**).
15. **Zixuan Yuan**, Hao Liu[†], Renjun Hu, Denghui Zhang, and Hui Xiong[†]. Self-Supervised Prototype Representation Learning for Event-Based Corporate Profiling. In Proceedings of the 35th AAAI Conference on Artificial Intelligence (**AAAI 2021, CCF-A**).
16. Shengming Zhang, Hao Zhong, **Zixuan Yuan**, and Hui Xiong[†]. Scalable Heterogeneous Graph Neural Networks for Predicting High-potential Early-stage Startups. The 27th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD 2021, CCF-A**).
17. Denghui Zhang, **Zixuan Yuan**, Yanchi Liu, Hao Liu, Zuohui Fu, Fuzhen Zhuang, Hui Xiong[†], and Haifeng Chen. Domain-Oriented Language Modeling with Adaptive Hybrid Masking

and Optimal Transport Alignment. The 27th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (**KDD 2021, CCF-A**).

18. **Zixuan Yuan***, Hao Liu^{*†}, Yanchi Liu, Denghui Zhang, Fei Yi, Nengjun Zhu, and Hui Xiong[†]. Spatio-Temporal Dual Graph Attention Network for Query-POI Matching. The 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval (**SIGIR 2020, CCF-A**).

Working Papers

1. Yifei Xu, Yu Cao, Haodong Hu, Pengyang Wang, **Zixuan Yuan**[†]. Evaluating Theory-of-Mind in Compressed Large Language Models. [Under review at **ARR 2026**].
2. Junren Xiao, **Zixuan Yuan**[†], Ruiyao Miao, Shiya Tsang. Disentangle, Gate, and Optimize: Cross Domain Transfer power by Multi Objective Bayesian Optimization. [Under review at **KDD 2026**].
3. Yidan Shen, Guangwei Zhang, Kaichen Zhang, Denghui Zhang, **Zixuan Yuan**[†]. From Easy Lies to Subtle Deception: Self-Paced Model-Adaptive Data Augmentation for Fake News Detection. [Under review at **IJCAI 2026**].
4. Yanshen Yi, Chao Wang, **Zixuan Yuan**, Qi Zhang, Zhenya Huang, Bing Yin. ASP: Dynamic Prompt Policy Learning with Tri-Agent Reasoning for Interpretable Event-Driven Financial Forecasting. [Under review at **IJCAI 2026**].
5. Wenwei Wu, Youhang Li, **Zixuan Yuan**[†]. Decouple to Distill: Structured Autoregressive Reasoning for Efficient Listwise Reranking. [Targeting on **NeurIPS 2026**].
6. **Zixuan Yuan***, Denghui Zhang*, Hao Zhong, Shengming Zhang AlphaVC: A Reinforcement Learning-Based Venture Capital Investment Strategy.
 - Accepted by Workshop on Information Technologies and Systems (**WITS 2022**) and INFORMS Workshop on Data Science (**DS 2022**).
 - Targeting on Management Science (**MS**).

TEACHING & MENTORSHIP

Sole Instructor, FinTech Thrust, HKUST(GZ)

- Postgraduate course: FTEC 5050, Machine Learning and Artificial Intelligence, Fall 2023 (4.92/5.0), Fall 2024 (4.86/5.0), Fall 2025 (4.90/5.0)

Sole Instructor, University Level, HKUST(GZ)

- Undergraduate course: UFUG 1102, Calculus I, Fall 2024 (4.47/5.0), Fall 2025 (4.59/5.0)

Sole Instructor, Rutgers Business School, Rutgers University

- Undergraduate course: 29:623:311, Production and Operations Management, Summer 2022
- Undergraduate course: 29:623:335, Data Warehousing & Data Mining, Spring 2023

Graduated RPg Students

- Ziyue Zhang, MPhil. (Fall 2023)
Research Area: AI-driven Startup Success Prediction
Placement: Quant at Beijing Pufeng Private Equity Fund Management Co., Ltd.
- Xin Qing, MPhil. (Fall 2023)
Research Area: Aligned LSTM Temporal Topic Modeling for Analyzing BCI Technology Commercialization
Placement: Risk analyst, Zangge Mining Investment (Chengdu) Co., Ltd.

Graduate Advisees

- Wenwei Wu, Ph.D. student (Fall 2024)
- Meishu Peng, Ph.D. student (Fall 2024), Prof. Yi Zhang as secondary supervisor
- Jian Chen, Ph.D. student (Fall 2024)
- Youhang Li, Ph.D. student (Fall 2025)
- Runhao Sha, MPhil. student (Fall 2023)
- Siyuan Li, MPhil. student (Fall 2024)
- Yidan Shen, MPhil. student (Fall 2024)
- Yuhan Xu, MPhil. student (Fall 2025)
- Binghao Liu, MPhil. student (Fall 2025)

Undergraduate Advisees

- Junkun Chen, Undergraduate student (Fall 2024)
- Shengxi Li, Undergraduate student (Fall 2025)
- Fai Ip TAM, Undergraduate student (Fall 2025)
- Tiantian Zhu, Undergraduate student (Fall 2025)

PROJECTS & GRANTS

1. **Research on Cross-scenario Decision Optimization and Collaborative Attribution based on Multimodal Big Data**, 2025-2027
PI, 300,000 RMB
NSFC Young Scientists Fund National Natural Science Foundation of China
2. **Domain Expert Knowledge Base Q&A: Architecture and Multi-Scenario Task Tuning**, 2025-2026
PI, 50,000 RMB
China Merchants Fund Management Co., Ltd.
3. **Research on Financial LLM Post-training**, 2024-2026
PI, 2,000,000 RMB
Dongguan Securities Co., Ltd.
4. **Research on AI-driven Corporate Compliance**, 2026-2027
PI, 300,000 RMB (Contract Pending)
HSBC
5. **AI-Powered Second Brain for Financial Data Management**, 2026-2028
Collaboration with Prof. Junming Liu from CityU
Co-PI, 10,000,000 RMB (Contract Pending)
CITIC Group

UNIVERSITY SERVICE

- Academic Advisor of Residential College, HKUST(GZ):
 - Mentored 4 undergraduate students: Junkun Chen, Shengxi Li, Fai Ip TAM, Tiantian Zhu, 2025-now.
- Undergraduate Admission Outreach:
 - Shanghai, HKUST(GZ), 2025.

- Freshman Orientation Week:
 - FTEC Info Session, Driving Financial Innovations with Cutting-Edge Technologies, 08-29-2025.
- RPG Recruitment (Thrust level):
 - Early admission (10 Ph.D. students) and department recruitment promotion, 2023–now.
- RPG Admission Outreach (Thrust level):
 - Virtual Information Session held by CEAIE, Introduction of PhD (FTEC) program, 06-26-2024.

SOCIAL SERVICE

- Deputy Secretary-General, *Guangdong Association for Science and Technology Finance Promotion*

KNOWLEDGE TRANSFER

- In January 2024, Guangzhou Lingyue Technology was founded in Nansha District.
- The company is dedicated to providing everyday investors with access to expert-level AI wealth management support.
- The company has partnered with 10 benchmark clients (top-ranked brokerages, banks, and fund management companies), covering use cases including knowledge-base Q&A, AI-driven stock investing, compliance inspection, best-practice sales scripts, and strategy recommendations.
- The company closed an angel round investment by the end of 2025.
- The company’s core product “SynWealth” has received strong positive feedback from customers. Our 2026 revenue is expected to exceed 7 million RMB.
- The company has secured a patent for our AI-driven investment advisory system, demonstrating its original innovation, industry adoption, and impact on applied FinTech research.

AWARDS & HONORS

- Dean’s Scholarship, Rutgers University Summer 2020
- Research Fellowship, Rutgers University Fall 2018 - Fall 2022
- Laura Bassi Scholarship, Editing Press Winter 2022

PATENTS

- Intelligent Robo-Advisory System. HKUST(GZ), Dongguan Securities. Chinese Patent. November 2025.
- Unsupervised Cross-Domain Data Augmentation for Long-Document Based Prediction and Explanation. Yada Zhu, **Zixuan Yuan**, David Cox, Anna Wanda Topol. US Patent. June 2025.

RESEARCH EXPERIENCE

IBM Research, NY, US

June 2021 - August 2021

Data Science Researcher (Internship)

- Developed an interpretable transformer-based language model to precisely evaluate the future performance and market volatility of publicly-traded companies based on their textual information (e.g., earning calls).

RavenPack, NY, US

November 2020 - February 2021

Data Science Researcher (Internship)

- Designed an explainable deep learning approach to perform the day-frequent trading in US and European stock markets by using NLP techniques to exploit market sentiments from corporate news events.

XQuant Company Ltd., Hangzhou, China

February 2020 - May 2020

Data Science Researcher (Internship)

- *Project I: Corporate Profiling*: Assessed the evolving operational status of corporates from its event sequence, and constructed event-based corporate profile via self-supervised prototype learning paradigm.
- *Project II: T+0 Algorithmic Trading*: Developed an online adaptive sequential modeling algorithm for tick-level intra-day trading in Chinese stock market.

Baidu Research, Beijing, China

May 2019 - August 2019

Data Science Researcher (Internship)

- Incorporated the incremental learning paradigm to shorten the application latency and adaptively study the evolving trip preference of Baidu Maps users.

Baidu Research, Beijing, China

May 2018 - August 2018

Data Science Researcher (Internship)

- Implemented the spatiotemporal graph learning methods to provide dynamic and personalized query-POI matching criterions for Baidu Maps users.

INVITED TALKS

- Deep Learning Enhanced Corporate Valuation. *Hong Kong University of Science and Technology*, Online, November 2022.
- Self-Supervised Prototype Representation Learning for Event-Based Corporate Profiling. *The session "High-dimensional Data Analytics for Modeling, Monitoring, and Control" at 2022 INFORMS annual conference*, Indianapolis, IN, US, October 2022.
- Exploiting the Power of Language Modeling for Improving the Earnings Call Analysis. *IBM Research*, Online, August 2021.
- Equity Investment Enhanced with News & Corporate Events. *RavenPack Research*, Online, October 2020.

PROFESSIONAL ACTIVITIES**Session Chair**

- The session AI-driven Business Analytics: New Advances and Applications at 2022 INFORMS annual meeting

Program Committee Member

- ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2023, 2024, 2025, 2026
- Association for the Advancement of Artificial Intelligence (AAAI), 2024, 2025, 2026
- International Joint Conference on Artificial Intelligence (IJCAI), 2025, 2026
- International Conference on Learning Representations (ICLR), 2024, 2025, 2026
- ACL Rolling Review (ARR), 2026
- ACM International Conference on AI in Finance (ICAIF), 2024, 2025
- The IEEE ICDM NeuRec-Workshop, 2023

Journal Reviewer

- IEEE Transactions on Knowledge and Data Engineering (TKDE), 2022, 2023, 2024, 2025
- ACM Transactions on Knowledge Discovery from Data (TKDD), 2022, 2023, 2024, 2025
- ACM Transactions on Intelligent Systems and Technology (TIST), 2024, 2025
- Frontiers of Computer Science, 2022, 2023, 2024, 2025
- Computer Communications, 2022, 2023, 2024, 2025
- World Wide Web, 2022, 2023, 2024, 2025